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Cervical cancer

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Introduction

Unlike ovarian cancer, for example, cervical cancer and its precancerous stage, cervical dysplasia, are "accessible" for easy examination due to their location and can thus potentially be identified earlier.

With a perfect cervical cancer screening program, there would be no more invasive cervical cancer, as SIL is a precancerous stage that can be detected using cervical cytology and colposcopy.
In a historical cohort study, 31 % of patients with HSIL diagnosed using punch biopsy alone were found to have developed invasive cervical cancer within 30 years. If the HSIL had been treated, only 0,7 % would have developed invasive cancer in the following 30 years [P06]. [P06]

According to the current 2014 WHO classification, the term CIN I–III (cervical intraepithelial neoplasia) has been replaced by the term Squamous Intraepithelial Lesion (SIL) [B08]:

- Low-grade Squamous Intraepithelial Lesion (LSIL) replaces CIN I
- High-grade Squamous Intraepithelial Lesion (HSIL) replaces CIN II and CIN III [B08]

Definitions

Invasive cervical cancer is either:

- Squamous cell: cervical HSIL (precancerous) --> invasive squamous cell carcinoma (80 %)
- Glandular epithelium: Adenocarcinoma in situ (precancerous) --> invasive adenocarcinoma (20 %)

Epidemiology

Cervical cancer is the most common female genital cancer worldwide.

The global age-standardized incidence rate is estimated at 14 per 100'000 per year, ranging from 2 (Palestine) to 75,9 (Malawi).

Its incidence is kept low in countries with high socioeconomic status thanks to early detection and treatment of precancerous stages.

In Germany, the age-standardized incidence according to the European population is 8,7 per 100'000 women per year (versus 16,5 for endometrial cancer and 119,6 for breast cancer) [P01].

In Switzerland, the incidence is above 4 and mortality is 2,5 to 1,5/100'000 per year. [P01]

Infection with oncogenic human papillomavirus (HPV) is an essential condition for the development of cervical dysplasia and thus cervical cancer (=> G10A).

- 99,7 % of cervical cancer (and 100 % of HSIL and adenocarcinoma in situ) is associated with persistent oncogenic HPV infection.
- HPV type 16 or 18 can be found in 71 % of all cervical cancer.
- A changing "HPV inoculum" as the result of multiple sexual partners correlates closely with the later occurrence of cervical cancer.

Other, non-causal, tumor promoters include:

- Smoking
- Vitamin deficiency (carotenoids, vitamin C, folic acid)
- Immunodeficiency (e. g. HIV, history of organ transplantation)
- Long-term use of ovulation inhibitors

Cervical cancer must be understood as a possible consequence of persistent infection with high-risk types of HPV:

- The most oncologically dangerous (as most common) high-risk HPVs are 16 and 18
- Rarer high-risk HPVs: 31, 33, 35, 39, 45, 51, 52, 55, 56, 58, 59, 66, 68, 70, 73, 82

The malignant degeneration of HPV-infected epithelial cells is associated with overexpression of the viral oncogenes E6 and E7. These inactivate tumor suppressor genes.

Early detection and HPV vaccination

--> see G11A

Clinical morphology

Histological types:

- Squamous cell carcinoma: Adenocarcinoma = 80 : 20

Neuroendocrine, clear cell and serous papillary cervical cancer are very rare histological types with poor prognosis.

Neuroendocrine, clear cell and serous papillary cervical cancer are very rare histological types with poor prognosis.

This area of the cervix is especially prone to oncogenic transformation, as human papillomaviruses can infect the proliferating cells of the basal epithelial layer here.

Spread, metastasis:

- Direct spread is often from the cervix to the parametrium, descending into the vagina. The bladder and rectum can also be invaded.
- Lymphogenic metastasis in various directions is typical (in the parametrium, the fossa obturatoria, along the internal/external iliac arteries and finally (pre)sacral and along the common iliac artery and paraaortic).
- Hematogenic metastasis is found in less than 5 % of patients.

With regard to direct spread, a distinction is drawn between

- the ventral parametrium (vesicouterine ligament)
- the lateral parametrium (paracervical tissue of the cardinal ligament)
- the dorsal parametrium (uterosacral ligament)

Clinical picture

- Most precancerous and early stages have few or no symptoms.
- Premenopause: metrorrhagia, contact bleeding (typically after intercourse).
- Premenopause and postmenopause: vaginal discharge (turbid to bloody, malodorous) and/or contact bleeding.
- Postmenopause: postmenopausal bleeding.
- If tumor growth is advanced, there may be pain in the pelvis, sacrum and lumbar region. Leg lymphedema is also a late symptom.
- Note regarding late stages: (i) hydronephrosis and subsequent uremia, (ii) arrosion bleeding, (iii) fistula.

Prognosis (factors)

Tumor stage is the most important prognosis factor in cervical cancer.

Thus, tumor spread (local invasion of parametrium, vagina, bladder, rectum), lymph node status (metastasized or not) and/or distant metastasis (organ metastasis) play a decisive role.

For tumors limited to the uterus (FIGO I), other characteristics are also important for the prognosis resp. treatment procedure:

- Depth of invasion (invasion of cervical stroma)
- Clinical tumor diameter (or tumor volume)
- Lymphatic invasion
- Blood vessel invasion

The presence of residual tumor after surgery (positive resection margins) is also important for prognosis. Incomplete resection has a negative impact.

Table 1: Stage-dependent 5-year survival rate (regardless of treatment type!)

FIGO stage	5-year survival
0	100 %
I	approx. 95 %
II	approx. 70 %
III	approx. 55 %
IV	approx. 20 %

Across all stages about 2/3 of the patients will be cured.

Diagnosis

Diagnosis is determined histologically (tissue biopsy)! Clinical and cytological suspicion, together with one or more of the following tests, help guide the practitioner:

- Speculum: The tumorous process can be identified with the naked eye at a relatively early stage if it is located on the ectocervix.
- Colposcopy of the ectocervix and vagina: Precancerous growths can be identified with a microscope (after application of acetic acid and if visible), and in general the location and spread of suspected/manifest findings.
- If changes are seen, punch biopsies of the suspicious sites must be taken. For ectocervical lesions visible macroscopically, a biopsy should take place for confirmation of the diagnosis, rather than a conization.
- Endocervical findings can only be identified with cervical curettage (outpatient, without general anesthetic) or conization.

The tumor spread is then determined (staging).

Locoregional spread:

- Clinical examination to assess local spread: microscopic vs. macroscopic expansion confined to the ectocervix, infiltration of vaginal walls.
- Examination under general anesthetic with bimanual rectovaginal examination: the best preoperative method for determining parametrial involvement: with experienced practitioners, this examination is more reliable than CT or MRI.
- Cystoscopy and rectoscoscopy (often carried out at the same time as parametrial examination under general anesthetic, to rule out infiltration of the bladder or rectum).
- MRI and expert transvaginal ultrasound (tumor size and additional parametrial examination, ureteral congestion).

Lymph node metastasis, organ metastasis: additional examinations then complete the staging information:

- PET-CT (lymph node and organ metastasis, "additional" information on tumor spread into neighboring organs).
- Surgical lymph node staging (lymphadenectomy) remains the best method for ruling out lymph node metastasis.
- Depending on the available diagnostic methods: chest x-ray (to rule out lung

metastasis); iv urography (ureteral congestion)

In a meta-analysis, PET-CT showed the highest success rate with regard to lymph node metastasis, although the false-negative rate is still 6 to 15 %, with a positive predictive value of 86 to 100 % [P09]. [P09]

Molecular biological evidence of HPV infection?

- is not indicated if HSIL or carcinoma has already been diagnosed. (HPV will be positive anyway, "what type it was" is no longer relevant).

Staging

Until 2018, cervical cancer staging was carried out clinically taking into account the following examinations: gynecological examination with colposcopy and biopsy sampling, examination under general anesthetic (with bimanual rectovaginal examination to assess parametrial infiltration), cystoscopy and colonoscopy, as well as renal ultrasound (hydronephrosis) and chest x-ray. The FIGO stage was then established and remained unchanged. Treatment was naturally adapted to any additional information that became available concerning the disease (e.g. lymph node involvement or organ metastasis), but the FIGO stage remained the same. It was a "clinical" staging.

Cervical cancer is widespread in developing countries where medical diagnosis is more limited. This is why staging was carried out on a clinical basis only, in order to initiate further treatment.

However, this led to major discrepancies between classification systems (TNM, FIGO). Diagnostic methods such as MRI and PET-CT were also increasingly common, leading to their inclusion in the new 2018 FIGO classification [B01]. The FIGO classification (2018) was adapted to/aligned with the TNM system. Following this, the FIGO stage could only be established after complete staging, taking into account all available methods. This means the classification includes both radiological findings from MRI or PET-CT and pathological findings following lymphadenectomy. The suffixes (r) and (p) are used to distinguish between lymph nodes suspected of being metastatic on imaging or proven to the metastatic pathologically (after biopsy) respectively. For example: FIGO IIIC(r) = Lymph node involvement based on imaging.

The new FIGO 2018 classification also included some other changes (e.g. the new FIGO stage IB3). This is relevant for patients' medical histories but also when reading expert literature. It is important to note which classification is being used so it can be looked up if necessary. [B01]

Table 2: FIGO 2018 and UICC 2010

FIGO 2018	UICC 2010	Description/Definition
I	T1	Cervical cancer confined to the uterus (extension to the corpus should be disregarded)

FIGO 2018	UICC 2010	Description/Definition
IA	T1a	Invasive cancer identified only microscopically, stromal invasion < 5 mm
IA1	T1a1	Stromal invasion < 3 mm in depth
IA2	T1a2	Stromal invasion ≥ 3 mm but < 5 mm in depth
IB	T1b	Invasive carcinoma with stromal invasion ≥ 5 mm or clinical lesions confined to the cervix
IB1	T1b1	Stromal invasion ≥ 5 mm in depth and < 2 cm at largest diameter
IB2	T1b2	Clinical lesion ≥ 2 and < 4 cm at largest diameter
IB3	.	Clinical lesion ≥ 4 cm at largest diameter
IIA	T2a	Invasion beyond the uterus with no parametrial involvement. Involvement of up to the upper two-thirds of the vagina
IIA1	T2a1	Clinical lesion < 4 cm with no parametrial involvement
IIA2	T2a2	Clinical lesion ≥ 4 cm with no parametrial involvement
IIB	T2b	Invasion beyond the uterus with parametrial involvement but no involvement of pelvic wall or lower third of the vagina
III	T3	Involvement of the lower third of the vagina and/or pelvic wall and/or renal congestion and/or non-functioning kidney and/or lymph node involvement
IIIA	T3a	Involvement of lower third of the vagina but no extension into the pelvic wall
IIIB	T3b	Extension into the pelvic wall and/or hydronephrosis or non-functioning kidney
IIIC	.	Involvement of pelvic and/or paraaortic lymph nodes, regardless of size and spread (with r and p notation)

FIGO 2018	UICC 2010	Description/Definition
IIIC1	N1	Involvement of pelvic lymph nodes (NB: IIIC1r = radiological diagnosis, IIIC1p = pathologic diagnosis)
IIIC2	M1	Involvement of paraaortic lymph nodes
IV	T4	Involvement of bladder, rectum, remote metastasis
IVA	T4	Invasion of mucosa of the bladder and/or rectum and extension beyond the true pelvis
IVB	T1-4, M1	Distant metastasis
in regional lymph nodes	in regional lymph nodes	Paracervical, parametrial, internal iliac, obturator canal, external iliac, common iliac, presacral and lateral sacral lymph nodes

Treatment

Cervical cancer treatment - microcarcinoma

Stage IA1 (lymph node involvement if L0: < 1 %, if L1: 8 %)

- LVSI (lymphovascular space invasion) is a risk factor for lymph node metastasis.
- L1 = presence of lymphatic invasion in tumor
- V1 = blood vessel invasion in tumor
- Stage IA1: lymph node involvement if L0 < 1 %, if L1 8 %
- Conization with endocervical curettage: if the excision of the tumor is complete (free margin), with no lymphatic system invasion, if patient desires children. Requires regular cytological and colposcopic check-ups.
- Simple hysterectomy: Can be recommended if the patient does not wish to have (more) children.
- Surgical lymph node staging should be discussed in stage IA1 L1.

Stage IA2 (8,3 % with lymph node metastasis even if L0)

- With 0 or 1 risk factors: extrafascial hysterectomy with surgical lymph node staging
- If the patient wishes to have children: Conization in sano with cervical curettage and surgical lymph node staging, trachelectomy with surgical lymph node staging
- IA2 with two risk factors (lymphatic system invasion, V1, G3): Radical hysterectomy type B with Staging lymphadenectomy, in case of pelvic lymph node involvement, paraaortic lymphadenectomy

Cervical cancer treatment - macrocarcinoma

Hysterectomy classification according to the FIGO 2018 Report (table 2)

This classification gives an overview of the indication and extent of resection in case of cervical cancer [P02]. [P02]

The traditional Wertheim hysterectomy and the Piver-Rutledge-Smith classification I–V from 1974 use imprecise anatomical terminology and do not take into account the autonomous pelvic nerve cords and the extent of lymphadenectomy. In addition, the extent of vaginal resection is overevaluated. This is why the Querleu-Morrow classification of radical hysterectomy is more frequently used [P07]. [P07]

This uses an international anatomical nomenclature:

- Paracervix: Lateral connective tissue of the cervix: includes cardinal ligament = Mackenrodt's ligament, lymphovascular tissue and paracolpium. The deep uterine vein runs between the cranial and caudal paracervix with the hypogastric plexus.
- Parametrium: Lymphovascular tissue around the cranial uterine vein of the paracervix.
- Vesicouterine ligament: paracystium

Type A:

- This uses an international anatomical nomenclature:
- Paracervix: Lateral connective tissue of the cervix: includes cardinal ligament = Mackenrodt's ligament, lymphovascular tissue and paracolpium. The deep uterine vein runs between the cranial and caudal paracervix with the hypogastric plexus.
- Parametrium: Lymphovascular tissue around the cranial uterine vein of the paracervix.
- Vesicouterine ligament: paracystium

Type B1:

- Partial resection of the vesicouterine ligament and uterosacral peritoneal fold of the rectouterine ligament.
- Ureter is unroofed and mobilized laterally, transection of the paracervix at the level of the ureteral tunnel.
- Caudal paracervix (under the deep uterine vein) not resected.
- At least 1 cm of the vagina is resected.

Type B2:

- B1 plus selective lymphadenectomy in lateral paracervical tissue

Type C:

- Transection of vesicouterine ligament at bladder and rectouterine ligament at the rectum.
- Ureter is completely mobilized.
- 15–20 mm vagina with paracolpium resected.

Type C1:

- With autonomic nerve preservation: The hypogastric nerves are separated in the

lateral rectouterine ligament and only the uterine branches with the rectouterine ligament are detached. The bladder branches of the pelvic plexus in the vesicovaginal ligament are preserved.

Type C2:

- The paracervix is completely resected.

Type D1:

- Resection of the entire paracervix at the pelvic sidewall, exposing the roots of the sciatic nerve.
- Gluteal inferior, internal pudendal and obturator vessels are resected.

Type D2:

- laterally extended endopelvic resection: whole entire paracervix with internal iliac vessels and fascial/muscular structures resected.

Lymphadenectomy

- Level I: internal and external iliac
- Level II: + common iliac with presacral lymph nodes
- Level III: + paraaortic lymph nodes caudal to inferior mesenteric vessel
- Level IV: + paraaortic lymph nodes infrarenal

Importance of surgical lymph node staging for cervical cancer > Stage IIB

Surgical lymph node staging by means of laparoscopic lymphadenectomy in cervical cancer that has progressed beyond IB is more sensitive than PET-CT with regard to lymph node metastasis. In 4–15 % of patients, paraaortic lymph node metastasis undetectable on PET-CT could be diagnosed using this method. These patients could potentially still be treated by expanding the radiation field. However, there are currently no prospective randomized studies on this [P09]. [P09]

Stage IB

- Radical hysterectomy (Wertheim or variants): The "traditional" Wertheim operation (extended radical hysterectomy) removes the uterus, parametrium, vaginal cuff, pelvic lymph nodes and if necessary other lymph nodes via laparotomy.
- The ovaries are generally preserved in order to help prevent menopausal side effects in premenopausal women.
- A similar operation can also be carried out vaginally (Schauta) or

laparoscopically.

Important information for surgical management:

- Laparoscopic radical hysterectomy is no longer seen as the standard procedure. Laparotomy became again the method of choice for radical hysterectomy in the late 2018.
- A prospective randomized study, the LACC Trial (Laparoscopic Approach to Cervical Cancer) [B02], in October 2018 showed an increased risk of mortality from cervical cancer if surgery was minimally invasive (laparoscopic or robotic) compared to open. The risk of cervical cancer mortality after three years was increased by a hazard ratio of 6.6. A retrospective SEER data analysis was published at the same time and confirmed a worse outcome after minimally invasive surgery (USA Cancer Registries, The Surveillance, Epidemiology, and End Results (SEER) Program). Over time, multiple data have been published by different medical societies and research groups showing the same overall trend. Patients should be informed about the LACC Trial before surgery. If they still decide on a laparoscopic approach, they should be informed of the data and the increased risk of relapse and death of cancer. This should be stated in the Informed Consent.
- Further studies to better understand the possible causes for this are currently being carried out. Creation of a vaginal cuff and avoidance of uterine manipulators could prevent the dissemination of cancer cells.
- If there is any intraoperative evidence of one or more lymph nodes metastasis, the radical hysterectomy should NOT be carried out, the operation should be aborted. The lymphadenectomy will then be completed, at least up to the inferior mesenteric vein (low paraaortic region), in order to define the radiation field - based on lymph node involvement - more precisely. If the lymph nodes are positive, chemoradiotherapy is indicated. Given the equally good outcomes, the aim is to reduce morbidity by not combining modalities in treatment. Radical hysterectomy combined with chemoradiotherapy is associated with up to 30 % secondary morbidity [B03].
  [B02][B03]

LACC Trial (Laparoscopic Approach to Cervical Cancer)

- Published by Ramirez et al. in the New England Journal of Medicine on October 31, 2018. [B02]
- Study comparing minimally invasive vs open radical hysterectomy for early-stage cervical cancer.

- Randomized, prospective, international multicenter Phase III study
- Non-inferiority trial
- Hypothesis: Laparoscopic or robotic radical hysterectomy is not inferior to abdominal radical hysterectomy with regard to disease-free survival (DFS) after 4.5 years. Secondary endpoints were recurrence and overall survival.
- The study was stopped prematurely (n=631 vs planned n=740) for safety reasons, as a significantly worse outcome was observed in the minimally invasive surgery arm.
- Results: 27 recurrences (of which 41 % in the fornix/pelvis) with 19 deaths in the minimally invasive group and 7 recurrences (of which 43 % in the fornix) with 3 deaths in the open group. Disease-free survival (DFS) after 4.5 years showed a difference of -10,6 % in the laparotomy group (DFS 86 % in the minimally invasive group vs 96,5 % in the open group). The hypothesis of non-inferiority was therefore not verified. The minimally-invasive procedure was significantly worse in terms of disease-specific mortality (HR 6.6; 95% CI 1.5–29.0) and locoregional recurrence (HR 4.3; 95 % CI 1.4–12.6).
- Summary: Minimally invasive radical hysterectomy is associated with a higher rate of recurrence and poorer overall survival and disease-specific survival than open radical hysterectomy.
  [B02]

In stages IB and IIA, particularly in young women and where permitted by the patient's overall condition, extended radical hysterectomy (traditionally: Wertheim operation) is carried out.

Variations of radical hysterectomy attempt to preserve the autonomic nerves (branches of the inferior hypogastric plexus and splanchnic nerves S2-S4).

Uterus-preserving radical trachelectomy is also admissible with small IB1 (< 2 cm diameter) in case of urgent desire to have children (= removal of cervix and partial resection of parametrium and pelvic lymph nodes).

Stage IIA

- Also removes a large part of the vagina (2 cm safety margin from tumor margin).
- In principle, beyond FIGO stage IIA, combined chemo-radiotherapy is indicated rather than surgery.

Stage IIB

- Combined chemo-radiotherapy.
- After chemo-radiotherapy, simple hysterectomy to reduce local relapse rate can be discussed (particularly if imaging shows possible residual tumor).

Adjuvant hysterectomy after chemo-radiotherapy reduced the local relapse rate, but not overall survival as shown in a randomized study.

Stages III and IV when localized

- Combined chemo-radiotherapy
- In stage IV with organ metastasis (primary metastatic cervical cancer) or in relapse (if local radiotherapy is no longer possible) or with persistent tumor after radio-chemotherapy
- First-line adjuvant systemic therapy with combined chemotherapy, +/- anti-angiogenic antibodies, + immunotherapy

Important information on systemic therapy and radiotherapy

With first-line adjuvant systemic therapy:

- For years, combined chemotherapy with cisplatin, paclitaxel and targeted therapy with bevacizumab (Avastin, anti-VEGF antibody, anti-angiogenic) (equivalent to the GOG 240 trial) [B04] was recommended as the standard first-line therapy. Since 2021, immunotherapy with pembrolizumab (anti-PD-1-antibody) has also been added!
- The combination of pembrolizumab plus cytostatics+/- bevacizumab is the new standard! [B05]
  [B04][B05]

Data from KEYNOTE-826, a Phase 3 randomized, placebo-controlled study

n=617 (Intention To Treat Population)

In patients with persistent, recurrent, or metastatic cervical cancer, a benefit for PFS (progression-free survival) and OS (overall survival) has been shown when the checkpoint inhibitor pembrolizumab (anti-PD-1-antibody) is combined with platin-based standard chemotherapy as first-line treatment. This benefit exists with and without additional treatment with an angiogenesis inhibitor (bevacizumab).

Median PFS was 10.4 months and median OS 24.4 months for pembrolizumab in the ITT population, versus 8.2 months and 16.3–16.5 months with placebo (hazard ratio [HR] for progress: 0.65; $p < 0.001$) (HR for all-cause mortality: 0.66; $p < 0.001$).

There was then stratification according to PD-L1 expression. The effect of the checkpoint inhibitor was predominantly observed in the PD-L1 positive group (Pd-L1 combined positive score ≥ 1 or ≥ 10). The significance of the checkpoint inhibitor for biomarker-negative tumors is therefore still unclear.

Combined chemo-radiotherapy

- For cervical cancer, a combination of chemotherapy and bimodal radiotherapy is used.

Chemotherapy: cisplatin, as a radiosensitizer, is administered without hospitalization and at a low dose on a weekly basis (alternative: carboplatin in case of kidney failure).

40 mg/m2 body surface i.v. weekly

This is carried out simultaneously with an intensive course of radiotherapy, comprising percutaneous high-voltage irradiation, followed by local contact irradiation.
Contact irradiation enables a biologically highly effective local dose to be applied.

Postoperative radiotherapy

- Exception, to be avoided wherever possible though careful indication of the operation.
- For example, there is an indication for postoperative radiotherapy if, despite all preoperative clarifications, affected lymph nodes or several other risk factors (parametrial involvement, lymphatic invasion, etc.) are found by surgery.

- Significant morbidity is expected with postoperative radiotherapy: leg lymphedema, radiation enteritis, fistula formation
- Diarrhea, blood or mucus in stool etc.
- Radiation damage to the bladder (mucosal ulceration or necrosis) --> painful urination, blood in urine, urinary tract obstruction, urge and bladder spasm, etc.
- In general, chemo-radiotherapy with cisplatin as a radiosensitizer is carried out in the same way as for primary therapy.

Cervical cancer and pregnancy

Cervical carcinomas and especially HSIL are often overlooked during pregnancy. HSIL is found in 0,33 % of all pregnant women, invasive cancer in 0,05 %.

Pregnancy does not appear to have any specific effects on the progression of cervical cancer. Treatment takes place depending on the stage, as for cases outside of pregnancy. It is currently not clear how far the start of treatment can be delayed:

- In the first trimester, the pregnancy is generally terminated and treatment started immediately.
- Later, depending on the findings and the expectations of the patient, it is possible to wait until the third trimester before carrying out a Cesarean section and (at the same time) radical surgery.

Managing relapse

If relapse is suspected, histological confirmation should be performed (biopsy, if necessary CT-guided).

In case of relapse, renewed staging should be done in order to determine the extent of the disease (chest/abdomen/pelvis CT, if necessary PET-CT, potentially MRI to determine local extent).

Treatment for local recurrence (no remote metastasis) depends on the primary therapy, location and spread of the recurrence and the patient's general condition. Therapy can range from radio(chemo)therapy to the full spectrum of surgical resection (up to exenteration).

- If radio-chemotherapy has not yet been carried out, this option is preferred. The chemotherapeutic agent of choice is cisplatin for combination with radiotherapy. However, carboplatin can be used alternatively in case of kidney failure or reduced tolerability.
- For localized recurrence with previously successful radio-chemotherapy, surgery aiming for R0 resection can be discussed with the patient. The patient should be informed about the possibility of an anterior and/or posterior exenteration (removal of the bladder/rectum). The procedure requires an interdisciplinary approach: gynecologists, abdominal surgeons, urologists, psycho-oncologists and potentially plastic surgeons are involved. The patient must be given comprehensive information on the potential life-altering measures. Exenteration requires extensive planning. For the patient, it means a permanent stoma (nephrostomy, colostomy), pelvic floor reconstruction, loss of sex organs (change in sexuality), frequent postoperative complications (wound healing disorders, empty pelvis syndrome, recurrent infections, etc.). The patient should reflect on the consequences of surgery in advance.
- If surgery is not a possibility (due to patient desire or in fragile patients), systemic therapy is also indicated for local recurrence/persistent tumor.

Systemic therapy is indicated for metastatic recurrence.

- Platin-based combination chemotherapy, +/- bevacizumab (monoclonal antibody against VEGF), with the recent addition of immunotherapy (pembrolizumab), is indicated as first-line therapy [B05].
- Monotherapy is generally recommended for patients with progression after

first-line therapy. However, monotherapy only shows low efficacy. More recent data show a better response with checkpoint inhibitors (immunotherapy) such as pembrolizumab, cemiplimab (phase III study) [B13], nivolumab including in combination with ipilimumab (CTLA-4 inhibitor). The use of immunotherapy is therefore growing. It shows a higher treatment response, which is sustained over time.
[B05][B13]

Follow-up care for cervical cancer

Follow-up care begins when primary treatment is completed. It comprises history-taking (asking about symptoms), physical examination (rectovaginal examination with speculum and cytology), medical advice, care and support. Follow-up care has two goals: identifying relapse promptly and supporting the patient in dealing with any side effects of treatment.

A follow-up appointment every three to four months for the first two to three years, then every six months for another two years is the international standard. Check-ups are then carried out on an annual basis [B07].

If any abnormal findings are identified during follow-up care, either from the patient's history (symptoms) or clinically during the gynecological examination, imaging is indicated in order to identify a locoregional or distant recurrence.

Imaging is thus based on symptoms and not carried out systematically at regular intervals. [B07]

Remember, that:

- current data suggests that earlier identification of recurrence (e. g. using imaging) is not associated with improved survival, but rather with a decrease in quality of life;
- cytological diagnosis only has limited informative value [P03]. Technical evaluation of a pap smear is complicated after successful radio(chemo)therapy due to extensive atrophic and radiogenic changes.
- physical/gynecological examination remains the best method for determining tumor recurrence. This appears to be able to detect the highest number of recurrences in both symptomatic and asymptomatic patients [P04].
  [P03][P04]

Patients require long-term care. The side effects of treatment are significant and include radiation cystitis, radiation vulvovaginitis, lymphedema, vaginal dryness, vaginal stenosis and vaginal fibrosis. Locoregional side effects (particularly of radio-chemotherapy) can also occur at a later point.

Sexual medicine care is essential, as sexuality is often affected after treatment (including with radical hysterectomy: sensory disorders, voiding dysfunction, vaginal shortening; with radio-chemotherapy: radiation dermatitis, telangiectasia with bleeding, fibrosis). Patients should therefore be advised in a specialized preventive consultation covering vulva/vagina care and vaginal dilatation. Regular stretching of the vagina with (silicon-) dilator (tube-shaped device, slowly increasing in size) can help prevent vaginal adhesion and thus preserve organ function.

Practice and Research

Increasingly, the sentinel lymph node biopsy is also used for staging purpose in cervical carcinoma. These sentinel lymph nodes are extensively processed with ultrastaging methods and examined for carcinoma cells. However, data on the safety of this staging procedure if used alone – meaning without systematic pelvic lymphadenectomy – is not yet mature (phase III studies still pending). The sentinel lymph node biopsy as staging procedure alone - is therefore not yet a standard in cervical cancer and should be carried out in the context of studies.

Literature

Nr.	Title	Author	Journal	Volume	Pages	Year
[B01]	Revised FIGO staging for carcinoma of the cervix uteri	Bhatla N, Berek J.S., Cuello Fredes M, Denny L.A., Grenman S., Karunaratne K., Kehoe S.T., Konishi I,Olawaiye A.B. et al.	Gynecolgoy and obstetrics			2019
[B02]	Laparoscopic Approach to Cervical Cancer	LaCC Trial, Ramirez et al.	NEJM			2018
[B03]	ABRAX Studie	Cibula D. et al.	ESMO			2020
[B04]	Improved Survival with Bevacizumab in Advanced Cervical Cancer	Tewari K.S. et al	N Engl J Med	370	734–743	2014
[B05]	Pembrolizumab for Persistent, Recurrent, or Metastatic Cervical Cancer	KEYNOTE 826	N Engl J Med	385	1856–1867	2021

Nr.	Title	Author	Journal	Volume	Pages	Year
[B06]	EMPOWER-Cervical 1/ GOG-3016/ ENGOT-cx9: Interim analysis of phase III trial of cemiplimab vs investigator's choice (IC) chemotherapy (chemo) in recurrent/ metastatic (R/ M) cervical carcinoma. Presented at the European Society of Medical Oncology 2021 Virtual Congress	Tewari K.S., Monk B.J., Vergote I et al.		Abstract VP4_2021	16–21	2021
[B07]	Follow-up for women after treatment for cervical cancer: a systematic review	Elit L. et al.	Gynecol Oncol	114(3)	528–35	2009

Nr.	Title	Author	Journal	Volume	Pages	Year
[B08]	WHO Classification of Tumours of Female Reproductive Organs	Kurman R.J., Carcangiu M.L., Herrington C.S., Young R. H. (eds.)	International Agency for Research on Cancer (IARC)			2014
[B09]	Revised FIGO staging for carcinoma of the cervix	Pecorelli S., Zigliani L., Odicino F.	Int J Gynaecol Obstet	105	107–8	2009
[B10]	Diagnosis and management of cervical cancer	Petignat P., Roy M.	BMJ	335	765–8	2007
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[P01]	Krebs in Deutschland 2020/06	Robert Koch-Institut (RKI)			
[P02]	FIGO CANCER REPORT 2018 Cancer of the cervix uteri	Bhatla N. et al.			
[P03]	Cervical Cancer Posttreatment Follow-up: Critical Analysis	Hillesheim I. et al.	Int J Gynecol Cancer	27(8)	1747-1752
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[P09]	Pretreatment retroperitoneal para-aortic lymph node staging in advanced cervical cancer	Smits R.M., Zusterzeel P.L.M., Bekkers R.L.M.	Int J Gynecol Cancer	23	973–983